EX. NO: 4

HILL CIPHER (2 X 2 MATRIX)

Aim:

To write a C program to implement the Hill Cipher, a polygraphic substitution technique, using a 2 x 2 key matrix for encryption.

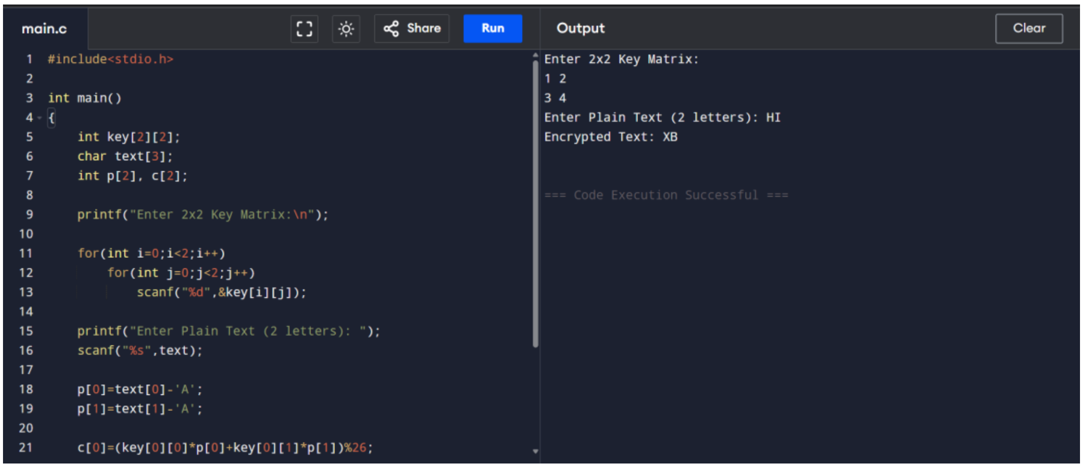
Algorithm:

1. Start the program.
2. Read the elements of the 2 x 2 key matrix from the user.
3. Read two letters of the plaintext and convert them into their numeric equivalents (A = 0, B = 1, ... , Z = 25).
4. Multiply the key matrix with the plaintext vector to obtain the cipher vector.
5. Reduce each element of the cipher vector modulo 26.
6. Convert the resulting numeric values back into their corresponding letters.
7. Display the encrypted text.
8. Stop the program.

Program:

```
#include <stdio.h>

int main() {
    int key[2][2];
    char text[3];
    int p[2], c[2];

    printf("Enter 2x2 Key Matrix:\n");
    for (int i = 0; i < 2; i++)
        for (int j = 0; j < 2; j++)
            scanf("%d", &key[i][j]);

    printf("Enter Plain Text (2 letters): ");
    scanf("%s", text);

    p[0] = text[0] - 'A';
    p[1] = text[1] - 'A';

    c[0] = (key[0][0] * p[0] + key[0][1] * p[1]) % 26;
    c[1] = (key[1][0] * p[0] + key[1][1] * p[1]) % 26;

    printf("Encrypted Text: ");
    printf("%c%c\n", c[0] + 'A', c[1] + 'A');
    return 0;
}
```

Output:



Result:

Thus a C program to implement the Hill Cipher using a 2 x 2 key matrix was written, executed and the output was verified successfully.